

Worksheet 1

Getting started

Modern electrical machines



Motors vary hugely in size, shape and technology.

Within a family of motors, for example DC motors, the characteristics of the machines are very similar - independent of their physical size and operating voltage.

This section looks at setting up the equipment. Then the testing can begin.

Over to you:

- 1) Connect the control unit to the mains power using the IEC connector. The quad 7-segment display and Power LED should light up.
- 2) Connect the dynamometer to the control unit with the 25 way D-type lead.
- 3) Make sure that the software applications and control unit driver are correctly installed.

When the control box is **not** connected to the PC using the USB lead, the box is in manual mode - the switches and knobs on the front panel determine the control box parameters.

Now you are ready to test the set up .

Set up a machine test

- 1) Line up the DC motor in the grooves of the dynamometer test bed.
It should slide in easily. You may have to undo the knurled nuts on the motor bed.

- 2) Make sure the packaging strap and sensor block are removed - see Unpacking in the reference section
- 3) Make sure that the rubber 'spider' which buffers the dynamometer and motor under test is in place. The system will work without this - it will just be noisy.
(On first use you might have to push hard!)
- 3) Make sure the motor and dynamometer couplings are aligned before pushing the motor home.
- 4) As you push home the DC motor into the dynamometer test bed you should hear the safety microswitches in the dynamometer cradle click.
- 5) Tighten the knurled lock nuts to make sure that the DC motor is secured in place.
- 6) Insert the balance and remove **both** protective plastic covers.

You are now ready to start experimenting with the equipment, in manual mode.